

Theory — Two-Camera Projective Geometry

Deriving the relationship between the image coordinates of a point $P(X, Y, Z)$ as seen by a static camera (camera 1) and a second camera (camera 2) at an oblique orientation.

1. Problem statement

Let a 3-D point $P = (X, Y, Z)^T$ lie within the field of view of two cameras:

- **Camera 1** is *static* (its coordinate frame is taken as the world frame).
- **Camera 2** is displaced from camera 1 by a rotation R and a translation t (an *oblique* orientation).

We wish to derive the mathematical relationship between P 's image coordinates in the two cameras,

$p_1 = (x_1, y_1, 1)^T$ and $p_2 = (x_2, y_2, 1)^T$ (homogeneous pixel coordinates).

The answer is the **epipolar constraint**, which ultimately takes the compact form

$$p_2^T F p_1 = 0$$

where F is the *fundamental matrix* (or, for calibrated cameras, the *essential matrix* E). This section derives it from first principles.

2. Assumptions and justification

#	Assumption	Justification
A1	Both cameras obey the pinhole model (no lens distortion).	The standard imaging model; distortion is removed in Step 1 by <code>cv2.calibrateCamera</code> , so the images are effectively pinhole projections.
A2	The two cameras are related by a rigid-body transformation (R, t).	Both cameras are static and mounted rigidly; a rigid motion (rotation + translation) is the most general mapping between two fixed Euclidean frames.
A3	The point P is in front of both cameras and at finite depth ($Z_1 > 0, Z_2 > 0$).	P is within the FoV of both cameras (given), so its depth along each optical axis is positive and finite.
A4	The scene is static — P does not move between the two views.	The cameras are simultaneous/static; otherwise the correspondence would not be a single 3-D point.
A5	The camera intrinsics K_1, K_2 are known (for the essential-matrix form).	Intrinsics are obtained from Step 1 calibration. If intrinsics are unknown, the weaker fundamental matrix still holds (projective, not metric).
A6	R is a proper rotation : $R^T R = I$, $\det(R) = +1$.	Rotations of physical rigid bodies lie in $SO(3)$.

3. Static parameters and variables

Symbol	Meaning	How determined
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$\mathbf{K}_1, \mathbf{K}_2$ (3x3)	Intrinsic matrices: $[[f_x, 0, c_x], [0, f_y, c_y], [0, 0, 1]]$	Camera calibration (Step 1). f_x, f_y = focal lengths in pixels; c_x, c_y = principal point.
$\mathbf{R}$ (3x3)	Rotation from camera-1 frame to camera-2 frame	Estimated from correspondences (see Sec. 7), or from a calibration rig.
$\mathbf{t}$ (3x1)	Translation from camera-1 frame to camera-2 frame (position of camera-1's center in camera-2's frame)	Same as $\mathbf{R}$; known up to scale for $\mathbf{E}$.
$\mathbf{E} = [\mathbf{t}]_x \mathbf{R}$ (3x3)	Essential matrix	Product of $\mathbf{t}$ and $\mathbf{R}$; recovered from $\mathbf{E}$ by SVD.
$\mathbf{F} = \mathbf{K}_2^{-T} \mathbf{E} \mathbf{K}_1^{-1}$ (3x3)	Fundamental matrix	From $\mathbf{E}$ and intrinsics, or directly from ≥ 8 correspondences (8-point algorithm).
$\mathbf{b} = \mathbf{t} $	Baseline (distance between the two optical centers)	Norm of $\mathbf{t}$; provides the absolute scale (ambiguous in $\mathbf{E}/\mathbf{F}$).
$\mathbf{e}_1, \mathbf{e}_2$	Epipoles (projection of one camera's center into the other's image)	$\mathbf{e}_1 = \mathbf{K}_1(-\mathbf{R}^T \mathbf{t})$, $\mathbf{e}_2 = \mathbf{K}_2 \mathbf{t}$ (up to scale).
$\mathbf{X}, \mathbf{X}_2$	$\mathbf{P}$ in camera-1 / camera-2 coordinates	Unknown (the 3-D point being observed).
$\mathbf{x}_1, \mathbf{x}_2$	Normalized image coords ($= \mathbf{X}/Z, = \mathbf{X}_2/Z_2$)	From pixels via $\mathbf{x} = \mathbf{K}^{-1} \mathbf{p}$.

Here $[\mathbf{t}]_x$ denotes the **skew-symmetric (cross-product) matrix** of $\mathbf{t}$,

$$[\mathbf{t}]_x = \begin{bmatrix} 0 & -t_3 & t_2 \\ t_3 & 0 & -t_1 \\ -t_2 & t_1 & 0 \end{bmatrix}$$

such that $[\mathbf{t}]_x \mathbf{v} = \mathbf{t} \times \mathbf{v}$ for any vector $\mathbf{v}$.

4. Single-camera pinhole model (recap)

For a camera with intrinsic matrix $\mathbf{K}$, a point $\mathbf{X} = (X, Y, Z)^T$ projects to pixel $\mathbf{p}$ by

$$\mathbf{Z} * \tilde{\mathbf{p}} = \mathbf{K} * \mathbf{X} \quad \Rightarrow \quad x = f_x * (X/Z) + c_x, \quad y = f_y * (Y/Z) + c_y$$

where $\tilde{\mathbf{p}} = (\mathbf{x}, \mathbf{y}, 1)^T$ is the homogeneous pixel vector. Dividing out Z , the **normalized** image point is

$$\hat{\mathbf{x}} = \mathbf{X} / Z = (X/Z, Y/Z, 1)^T$$

so that $\mathbf{p} = \mathbf{K} \hat{\mathbf{x}}$. This is the same perspective-projection equation used for the single-camera measurement in Step 2.

5. Two-camera setup (relative pose)

Let camera 1's frame be the world frame, so for $\mathbf{P} = \mathbf{X}$:

$$\mathbf{X}_1 = \mathbf{X} \quad (\text{camera-1 coordinates})$$

$$\mathbf{X}_2 = \mathbf{R} \mathbf{X}_1 + \mathbf{t} \quad (\text{camera-2 coordinates})$$

Using the normalized projections,

$$X_1 = Z_1 x_1 \quad \text{where } x_1 = X_1/Z_1$$

$$X_2 = Z_2 x_2 \quad \text{where } x_2 = X_2/Z_2$$

Substituting into the relative-pose equation:

$$Z_2 x_2 = Z_1 R x_1 + t \quad (1)$$

6. Derivation of the epipolar constraint

Starting from Eq. (1), premultiply both sides by the skew-symmetric matrix $[t]_x$:

$$[t]_x (Z_2 x_2) = Z_1 [t]_x R x_1 + [t]_x t$$

Since $[t]_x t = t \times t = 0$, this simplifies to

$$[t]_x (Z_2 x_2) = Z_1 [t]_x R x_1 \quad (2)$$

Now take the inner product of both sides with x_2 (left-multiply by x_2^T):

$$x_2^T [t]_x (Z_2 x_2) = Z_1 x_2^T [t]_x R x_1$$

The left-hand side is $Z_2 * x_2^T (t \times x_2)$. Because the cross product $t \times x_2$ is **perpendicular** to x_2 , the dot product vanishes:

$$x_2^T (t \times x_2) = 0 \quad \Rightarrow \quad \text{left-hand side} = 0$$

Therefore

$$x_2^T [t]_x R x_1 = 0 \quad (\text{normalized coordinates})$$

Defining the **essential matrix**

$$E = [t]_x R$$

gives the compact form

$$x_2^T E x_1 = 0 \quad (\text{epipolar constraint, normalized coordinates})$$

7. From normalized to pixel coordinates — the fundamental matrix

Using $x_1 = K_1^{-1} p_1$ and $x_2 = K_2^{-1} p_2$:

$$(K_2^{-1} p_2)^T E (K_1^{-1} p_1) = 0$$

$$p_2^T (K_2^{-T} E K_1^{-1}) p_1 = 0$$

Defining the **fundamental matrix**

$$F = K_2^{-T} E K_1^{-1} = K_2^{-T} [t]_x R K_1^{-1}$$

yields the pixel-domain relationship

$$p_2^T F p_1 = 0$$

For the special (common) case of identical cameras $K_1 = K_2 = K$, this is simply

$$F = K^{-T} E K^{-1}.$$

8. Geometric interpretation

- **Epipolar line.** For a point p_1 in image 1, the possible locations of the corresponding point in image 2 are constrained to the line $l_2 = F p_1$ (in homogeneous coordinates). The correspondence p_2 must satisfy $p_2^T l_2 = 0$, i.e. p_2 lies on l_2 . This is the one-dimensional constraint that turns 2-D point matching into a line search.
- **Epipoles.** The epipole e_2 is the projection of camera 1's optical center into image 2, and e_1 is the projection of camera 2's center into image 1. They are the (right and left) null vectors of F :

$$F e_1 = 0, \quad F^T e_2 = 0$$

with $e_1 = K_1(-R^T t)$ and $e_2 = K_2 t$. All epipolar lines in image 2 pass through e_2 .

- **Degrees of freedom.**

- E has 5 dof: 3 for R and 2 for the direction of t (t is known only up to scale).
- F has 7 dof: a 3×3 matrix of rank 2, defined up to scale (9 - 1 scale - 1 rank constraint).

9. Determining the parameters

Intrinsics (K_1, K_2). Obtained once by calibration (Step 1): the camera matrix gives f_x, f_y, c_x, c_y , and the distortion coefficients correct the images.

Fundamental matrix F (from image correspondences only). Given $N \geq 8$ point correspondences ($p_{1_i} \leftrightarrow p_{2_i}$), each pair contributes one linear equation in the 9 entries of F (writing $p_2^T F p_1 = 0$ as a dot product $(p_2 \times p_1)^T \text{vec}(F) = 0$). Stack N equations and solve the resulting linear system (the **8-point algorithm**); enforce $\text{rank}(F) = 2$ by setting the smallest singular value to zero. (Normalize point coordinates first for numerical stability.)

Essential matrix E and the relative pose (R, t). Once F is known and the cameras are calibrated,

$$E = K_2^T F K_1$$

Decompose E by SVD: $E = U \text{diag}(s, s, 0) V^T$ (enforce the two equal singular values). The rotation and translation are recovered from U and V :

$$R = U W V^T \quad (\text{or } U W^T V^T), \quad t = \pm u_3$$

$$\text{where } W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and } u_3 = \text{third column of } U$$

This yields **four** candidate (R, t) pairs; the correct one is selected by the **cheirality** test — triangulating P and keeping the solution for which P lies **in front of both** cameras ($Z_1 > 0$ and $Z_2 > 0$).

Absolute scale (baseline). E and F determine R and the direction of t , but not $|t|$ — the geometry is defined up to an overall scale. The baseline $b = |t|$ must be fixed by a known physical measurement (e.g., the measured object distance in Step 3, or a calibration object of known size).

10. Connection to the single-camera measurement (Step 2)

The Step-2 measurement $L = (dx / fx) * Z$ is the **fronto-parallel special case** of this geometry. If camera 2 differs from camera 1 by a pure horizontal translation ($R = I, t = (b, 0, 0)^T$ — a **rectified stereo**

pair), the epipolar constraint reduces to horizontal epipolar lines, and the horizontal **disparity** $d = x_1 - x_2$ of a point P gives its depth directly:

$$Z = f_x * b / d , \quad X = (x_1 - c_x) * Z / f_x , \quad Y = (y_1 - c_y) * Z / f_y$$

Thus the two-camera relationship derived here is the general setting of which the single-camera perspective-projection measurement is a limiting case.